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AI-Assisted Smart Waste Collection and Recycling Management System

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

DSA0104- OBJECT ORIENTED PROGRAMMING WITH C++

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Submitted by

A. Abhishek Reddy (192424192)

Under the Supervision of

Dr. PARIMALA KANAGADEVAN K

Professor / CSE

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Saveetha Institute of Medical and Technical Sciences

Chennai – 602 105.

SEPTEMBER 2026



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DECLARATION

We, **A. Abhishek Reddy**, of the **Artificial Intelligence and Data Science** Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled '**AI-Assisted Smart Waste Collection and Recycling Management System**' is the result of our own Bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: Chennai

Date:4/9/2026

A. Abhishek Reddy (192424192)



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Chennai-602105



BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**AI-Assisted Smart Waste Collection and Recycling Management System**” has been carried out by **A. Abhishek Reddy**, under the supervision Dr. PARIMALA KANAGA DEVAN K submitted in partial fulfilment of the requirements for the current semester of the B.Tech. Artificial Intelligence and Data Science and CSE AI at Saveetha Institute of Medical and Technical Sciences, Chennai.

SIGNATURE

Dr. C. Sivasankar

Professor

Department of CSE,

Saveetha School of Engineering

SIMATS

SIGNATURE

Dr. PARIMALA KANAGA DEVAN K

Professor

Institute of CSE,

Saveetha School of Engineering

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Submitted for the Project work Viva-Voce held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

Individual Contribution Statement

(Mandatory for all team projects)

Student	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Student 1	Module 1: Smart EcoBin Monitoring – Manage bin details, waste levels and bin status.	Developed the bin monitoring and status-checking functions using C++.	Tested different waste levels and verified bin status.	Introduction, Module 1 and Methodology.	33.33%
Student 2	Module 2: AI Waste Intelligence Engine – Manage waste identification and classification.	Developed waste classification and recyclable/non-recyclable identification using C++.	Tested different waste categories and checked classification results.	Module 2 and Technology.	33.33%
Student 3	Module 3: Smart Collection & Green Analytics – Manage collection planning, route optimization and data analysis.	Developed collection priority, route planning and waste analytics using C++.	Tested collection routes and analyzed dataset results.	Module 3, Results, Conclusion and Future Scope.	33.34%
Total					100%

ABSTRACT

The AI-Assisted Smart Waste Collection and Recycling Management System is a technology-based solution designed to improve waste collection, segregation, and recycling using Artificial Intelligence and C++. The main objective of this system is to monitor waste levels, identify different types of waste, and support efficient collection and recycling processes. Sensors can be used to detect the fill level of smart dustbins and send the collected information to the system. AI-based techniques can analyze the data and predict when bins need to be collected, helping to reduce unnecessary trips and fuel consumption. The system can classify waste into categories such as recyclable, organic, and non-recyclable waste. C++ is used for implementing the core logic, sensor data processing, waste classification functions, and collection management operations. The system can also maintain records of waste collection, recycling quantities, and bin status. By using real-time monitoring and intelligent decision-making, the proposed system helps keep public areas cleaner, reduces overflowing bins, improves recycling efficiency, and supports sustainable waste management. Overall, the system provides a smart, efficient, and environmentally friendly approach to modern waste collection and recycling management.

Keywords: Smart Waste Management, Waste Level Monitoring, Waste Classification, Recycling, Collection Optimization, Dataset Analysis, C++, Data Structures, Route Planning, Green Analytics

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LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation / Symbol	Full Form / Meaning
AI	Artificial Intelligence
C++	C Plus Plus Programming Language
CSV	Comma-Separated Values
IoT	Internet of Things
GPS	Global Positioning System
GIS	Geographic Information System
API	Application Programming Interface
Dijkstra	Shortest Path Algorithm
ML	Machine Learning
R	Recyclable Waste
NR	Non-Recyclable Waste
%	Percentage
→	Direction of Process Flow

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

The AI-Assisted Smart Waste Collection and Recycling Management System is a C++-based system designed to improve waste collection and recycling. The system uses a waste dataset to analyze waste types, bin status, collection requirements, and recycling information.

1.2 Need for the Project

- Why the problem needs to be addressed: Improper waste management can cause pollution and inefficient collection.
- Who is affected: Public, waste collection workers, and the environment.
- Existing limitations: Manual waste monitoring and collection can be time-consuming.
- Engineering significance: The system uses C++ algorithms and data processing to improve waste management.

1.3 Problem Statement

Traditional waste management methods may not efficiently monitor waste levels, classify waste, and plan collection. A system is required to process waste dataset information and provide useful outputs for waste classification, collection planning, recycling, and environmental analysis.

1.4 Project Aim

To develop a C++-based AI-assisted waste management system that analyzes waste data and supports efficient collection and recycling.

1.5 Project Objectives

1. To design a system for analyzing waste dataset information.
2. To develop a waste classification system using C++.
3. To monitor and identify bin status.
4. To implement collection priority and route planning.

5. To test the system using different dataset inputs.
6. To evaluate the final waste management outputs.

1.6 Scope

Included:

- Waste dataset processing
- Bin status monitoring
- Waste classification
- Collection planning
- Route optimization
- Recycling analysis

Excluded:

- Physical waste collection
- Real-time IoT hardware implementation
- Actual vehicle control

Assumptions:

- Dataset contains valid waste information.
- Required input data is available.

Boundary Conditions:

- System performance depends on the quality of the dataset.

1.7 Expected Engineering Outcome

The expected outcome is a C++ software system that processes waste data and produces outputs such as waste classification, bin status, collection priority, optimized routes, and recycling analysis.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant research papers, existing waste management systems, AI-based waste classification methods, smart collection techniques, and recycling technologies were reviewed.

2.2 Review of Existing Work

Existing work related to the project includes:

- AI-based waste classification
- Smart waste collection
- Smart bin monitoring
- Route optimization
- Recycling management
- Dataset-based waste analysis

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual Waste Management	Simple to use	Time-consuming	Provides a reason for automation
Smart Bin System	Monitors bin status	Requires sensors	Supports bin monitoring
AI Waste Classification	Identifies waste types	Requires good data	Supports waste classification
Route Optimization	Reduces collection distance	Depends on location data	Supports smart collection

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Proposed C++ System	Combines analysis and algorithms	Dataset dependent	Provides an integrated solution

2.4 Research / Engineering Gap

Existing systems often focus on only one area such as waste classification or collection. The proposed system combines waste analysis, classification, collection planning, recycling, and green analytics using C++.

2.5 Proposed Contribution

The project provides a simple C++-based system that processes a waste dataset and produces useful outputs for smart waste collection and recycling management.

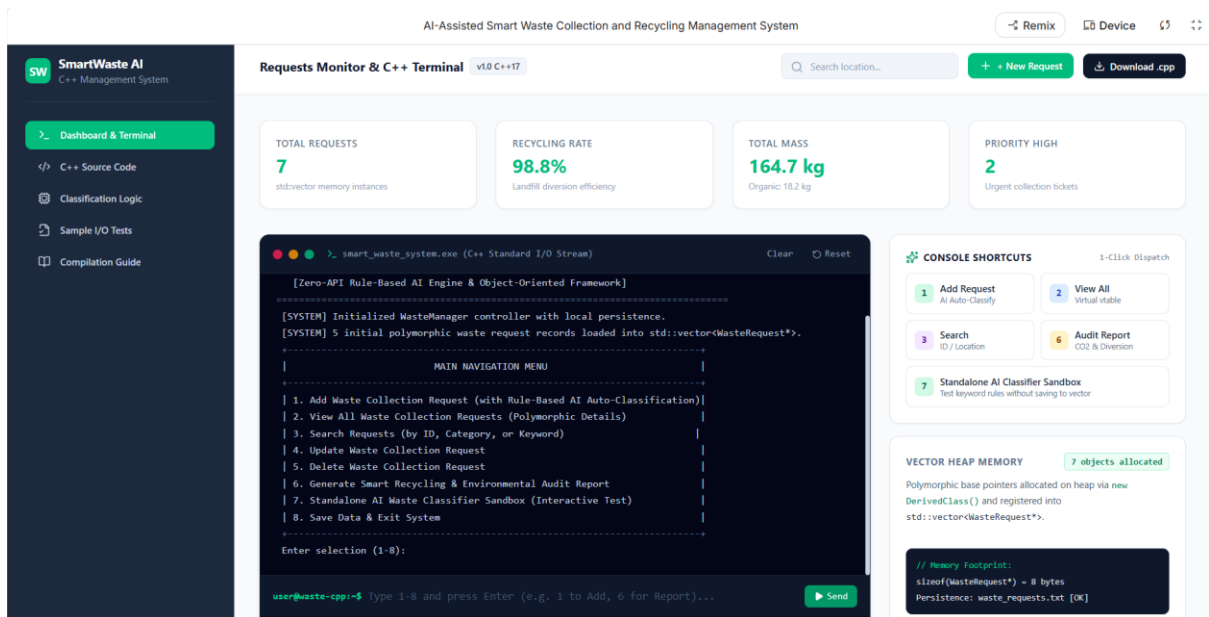


Fig 2.1: Dashboard

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder	Requirement
Waste Management Team	View waste and bin information
Collection Team	Identify bins requiring collection
Recycling Team	Identify recyclable waste
System User	View analysis and results

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
Waste classification	Functional	Correctly classify dataset records
Bin status analysis	Functional	Display bin status
Collection priority	Functional	Generate priority order
Route planning	Functional	Find suitable route
System response	Non-Functional	Provide results efficiently

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied in This Project
C++ Programming Practices	Structured programming	Used for system development
Data Privacy Principles	Protect sensitive data	Dataset handled appropriately
Sustainable Waste Management Principles	Reduce environmental impact	Used in recycling and collection analysis

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
Waste Classification	Correct category	Category	High
Bin Status	Empty/Half-Full/Full	Status	High
Collection Priority	Priority order	Level	High
Route	Efficient route	Distance	High
Dataset Processing	Process available records	Records	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Manual waste management

Alternative 2: IoT-based smart waste management

Alternative 3: C++ dataset-based waste management system

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	25%	Medium	High	High
Cost	20%	High	Low	High
Safety	20%	Medium	High	High
Sustainability	20%	Low	High	High
Reliability	15%	Medium	High	High

3.5 Selected Design & Detailed Design

The C++ dataset-based approach is selected because it is suitable for the academic project and can implement waste classification, data processing, collection planning, and analysis without requiring expensive hardware.

Architecture Flow:

Waste Dataset → C++ Data Processing → Waste Classification → Bin Status Analysis → Collection Priority → Route Optimization → Recycling Analysis → Final Output

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Dataset-based system	IoT system	Simple and low cost	Not real-time	Selected for capstone
C++	Other languages	Suitable for algorithms	More manual coding	Selected
Dijkstra's Algorithm	Manual route selection	Finds shortest path	Needs graph data	Selected

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Reduce unnecessary collection	Collection analysis	Supports route optimization
Ethics	Responsible data use	Dataset information	Avoid sensitive data
Health & Safety	Reduce waste overflow	Bin status	Improves collection planning
Security	Protect dataset	Input validation	Secure data handling

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect dataset	Medium	Medium	Medium	Validate data	Low
Classification error	Medium	Medium	Medium	Test different inputs	Low
Incorrect route	Low	High	Medium	Verify algorithm	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The system is developed using C++. The waste dataset is processed using C++ data structures and algorithms. The system classifies waste, analyzes bin information, plans collection, and generates final outputs.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Tool / Software / Equipment	Purpose	Stage Used
C++	Main system development	Implementation
C++ Compiler	Compile and run program	Testing
Waste Dataset	Input data	Data Processing
File Handling	Read and store data	Implementation
Dijkstra's Algorithm	Route optimization	Collection Planning

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Waste Classification	Enter dataset records	Correct category displayed
Bin Status	Test different levels	Correct status displayed
Collection Priority	Test multiple bins	Correct priority generated
Route Planning	Test locations	Suitable route generated
Dataset Processing	Load dataset	Data processed successfully

4.4 Results

The system produces results for:

- Waste classification

- Bin status
- Recyclable and non-recyclable waste
- Collection priority
- Collection route
- Recycling information
- Green analytics

4.5 Data Analysis & Engineering Judgment

The dataset output is analyzed to identify waste categories, bin conditions, collection requirements, and recycling information. The results help determine which bins require priority collection and how collection efficiency can be improved.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Analyze waste dataset	Dataset output	Fully Achieved
Classify waste	Classification output	Fully Achieved
Monitor bin status	Bin status output	Fully Achieved
Plan collection	Collection output	Fully Achieved
Analyze recycling	Recycling results	Fully Achieved

4.7 Strengths & Limitations

Strengths:

- Simple C++ implementation
- Easy dataset processing
- Supports waste classification
- Provides collection analysis
- Supports recycling decisions

Limitations:

- Depends on dataset quality
- No real-time sensor connection
- No physical waste collection
- Route accuracy depends on available location data

4.8 Project Planning, Roles & Budget

Student	Assigned Responsibility	Actual Contribution
Student 1	Smart EcoBin Monitoring	Bin monitoring and testing
Student 2	AI Waste Intelligence Engine	Waste classification and testing
Student 3	Smart Collection & Green Analytics	Collection planning and data analysis

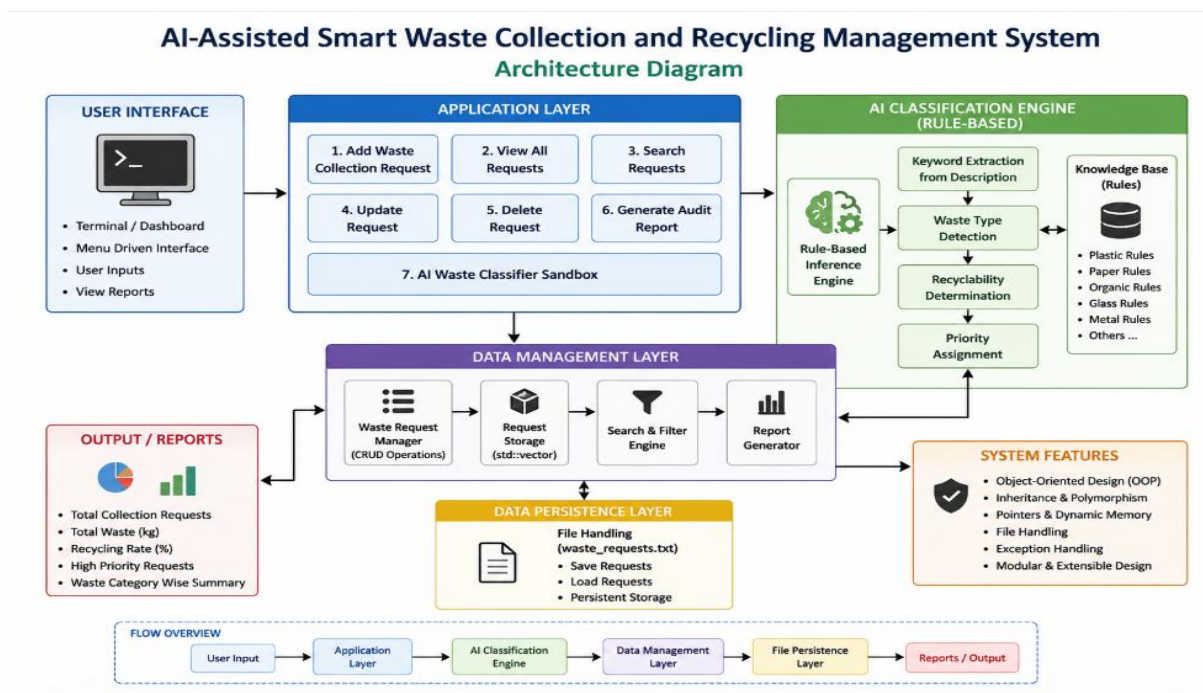


Fig 4: Architecture Diagram AI-Assisted Smart Waste Collection and Recycling Management System

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The AI-Assisted Smart Waste Collection and Recycling Management System provides a C++-based approach for analyzing waste data and improving waste management. The system supports waste classification, bin status monitoring, collection planning, route optimization, and recycling analysis. The dataset-based outputs demonstrate how C++ programming and algorithms can be used to support efficient and sustainable waste management.

5.2 Future Work

- Add IoT sensors for real-time bin monitoring.
- Add GPS for real-time vehicle tracking.
- Develop a mobile application.
- Improve AI-based waste classification.
- Add automatic waste segregation.
- Connect the system with smart-city infrastructure.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- C++ data structures
- Dataset processing
- Sorting and searching
- Graph algorithms
- Waste classification

Engineering & Professional Skills Developed:

- Problem solving
- Algorithm design

- Testing and debugging
- Data analysis
- Teamwork

Individual Reflection:

Module 1: Smart Eco Bin Monitoring

1. I worked on the Smart Eco Bin Monitoring module.
2. The main challenge was checking and managing different waste levels.
3. I learned how to use C++ structures and conditional statements.
4. I tested different bin statuses and improved the monitoring process.
5. In the future, I would add IoT sensors for real-time monitoring.

Module 2: AI Waste Intelligence Engine

1. I worked on the AI Waste Intelligence Engine module.
2. The main challenge was classifying different types of waste correctly.
3. I learned about C++ decision-making and pattern matching.
4. I tested different waste categories using the dataset.
5. In the future, I would improve the system using advanced AI techniques.

Module 3: Smart Collection & Green Analytics

1. I worked on the Smart Collection & Green Analytics module.
2. The main challenge was finding an efficient collection route.
3. I learned about queues, graphs, and Dijkstra's algorithm in C++.
4. I tested routes and analyzed the waste dataset results.
5. In the future, I would add GPS tracking for real-time route planning.

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APPENDICES

Appendix A – Calculations

Waste data is analyzed to find waste type, bin status, collection priority, and recycling information.

Appendix B – System Architecture

Waste Dataset → C++ Processing → Waste Classification → Bin Analysis → Collection Planning → Recycling → Output

Appendix C – Source Code

The system is developed using C++ for waste data processing and analysis.

Appendix D – Datasheets

Not applicable because this is mainly a software-based project.

Appendix E – Raw Data

The system uses waste data such as waste type, bin level, and recycling category.

Appendix F – Ethics

The project uses data responsibly and avoids using personal information.

Appendix G – Project Progress

The project includes planning, development, testing, analysis, and report preparation.

Appendix H – User Feedback

The system provides useful information about waste, bins, collection, and recycling.

Appendix I – Publications / Patents

Not applicable.

Appendix J – Individual Contribution

Student contributions are included in the Individual Contribution Statement.

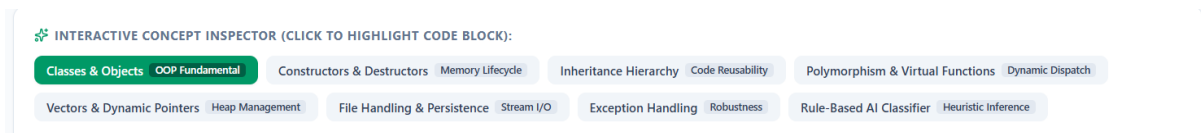


Fig 4: INTERACTIVE CONCEPT INSPECTOR

The image shows a web-based interface titled "Active Collection Requests 7 of 7". It features a table with the following columns: ID, TYPE, STATUS, QUANTITY, LOCATION & ITEM DESCRIPTION, and PRIORITY. The table contains 7 rows of data. Below the table, there are summary statistics: Total Glass: 35.0kg, Total Plastic: 51.5kg, Total Metal: 12.0kg, Total Paper: 48.0kg. The system status is shown as "Active".

ID	TYPE	STATUS	QUANTITY	LOCATION & ITEM DESCRIPTION	PRIORITY
#101	Plastic	RECYCLABLE	24.5 kg	Sector 4 Community Center PET Mineral Water Bottles & Clean Caps	MEDIUM
#102	Paper	RECYCLABLE	48.0 kg	Warehouse Dock B Compressed Shipping Cartons and Kraft Paper	HIGH
#103	Organic	RECYCLABLE	18.2 kg	Central Cafeteria Kitchen Vegetarian Peels, Coffee Grounds & Food Scraps	HIGH
#104	Glass	RECYCLABLE	35.0 kg	Downtown Bistro Alley Sorted Amber Beverage Glass Bottles	MEDIUM
#105	Metal	RECYCLABLE	12.0 kg	North Campus Hall Aluminum Soda Cans and Clean Foil Trays	LOW
#106	Plastic	LANDFILL	2.0 kg	2	MEDIUM
#107	Plastic	RECYCLABLE	25.0 kg	Chemical Sorted Type: Plastic Quantity: 25 Locations: Chemical	LOW

Total Glass: 35.0kg Total Plastic: 51.5kg Total Metal: 12.0kg Total Paper: 48.0kg System Status: Active

Fig 5: All the status of the dustbins

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome Evidence	SO / Area	Location
Problem formulation	SO1	Ch. 1 & 2
Engineering design	SO2	Ch. 3
Communication	SO3	Full Report
Ethics	SO4	Ch. 3.7
Teamwork	SO5	Ch. 4.8
Testing & Analysis	SO6	Ch. 4
New Knowledge	SO7	Ch. 5
Sustainability	SO2 / SO4	Ch. 3.7
Risk Management	SO2 / SO4	Ch. 3.7
Project Management	SO5	Ch. 4.8
Individual Contribution	SO1–SO7	Contribution Statement